



UNIVERSITY OF JOHANNESBURG

COLLEGE OF BUSINESS AND ECONOMICS

MASTERS IN FINANCIAL ENGINEERING

Block chain Assignment 2

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Assignment Report

Introduction

The purpose of this assignment is to implement a simple blockchain structure and demonstrate how cryptographic hashing can be used to protect the integrity of recorded transactions. The main problem is to ensure that once information has been recorded in a block, any change to that information can be detected. To achieve this, the implementation uses a `Block` class containing the main block fields, canonical JSON serialisation, and SHA-256 hashing. The code also creates a genesis block, links a subsequent block to it, and performs a tamper demonstration by changing a transaction and checking whether the stored hash is still valid. The logged output and assertions provide evidence that the code executes correctly and that tampering is detected.

From a Financial Engineering perspective, the exercise is relevant because financial systems depend on the integrity and reliability of transaction and market data. A change to a transaction like price, payment, and other financial record can affect calculations, valuations and audit results.

Question 1

I implemented a `Block` class to represent an individual block in the blockchain. The class contains an index, timestamp, a list of transactions, the previous block's hash, a nonce, and the block's own hash. I used the provided class template as a starting point and adapted it to the requirements of the question. The hash is calculated automatically from the block's information using SHA-256 rather than being manually entered.

During implementation, I tested the class by creating a sample block and checking that its stored hash matched the hash produced when the block was recomputed. This helped confirm that the block was correctly storing its required information and that the hash was connected to the contents of the block. This structure was then used in the following questions to demonstrate hash commitment and tamper detection.

Question 2

I created a second block containing a transaction with details such as the sender, recipient, amount, currency, date, and description. The material fields of the block were collected using `payload_for_hash()` and converted into a consistent JSON format using `canonical_dumps()`. The keys are sorted and unnecessary spaces are removed so that the same block data always produces the same representation. This JSON representation is then passed through SHA-256 to produce the block hash.

I then compared the calculated hash with the hash stored in the block. The assertion confirmed that both hashes were identical, showing that the hash computation includes the block's material fields. This demonstrates that the transaction information and other block details are committed to by the hash.

Question 3

I created the genesis block as the first block in the blockchain, giving it an index of 0, an empty transaction list, a nonce of 0, and a previous hash consisting of 64 zeros. I used a fixed timestamp so that the hash would be reproducible when testing the code. I then checked that the genesis block contained the correct values and that its stored hash matched the recomputed hash.

I then created a second block containing a sample transaction. The new block's index was set to 1, and its `previous_hash` was set equal to the hash of the genesis block. This links the two blocks together. The assertions confirmed that Block 1 was correctly linked to the genesis block and that its hash was also valid.

Question 4

To demonstrate tamper evidence, I made a copy of Block 1 and changed the transaction amount from 100 to 200. The original stored hash was kept unchanged, while a new hash was calculated using the altered transaction. The two hashes were different, showing that changing the transaction also changes the block's hash.

I then used the hash validation function to check the modified block. The validation returned `False`, confirming that the stored hash no longer matched the block's contents. The original Block 1 remained valid because `deepcopy` was used, showing that the tampering was detected without changing the original block.

1 GitHub Repository

The source code for this assignment is available on GitHub:

<https://github.com/Siphosethu19D/BlockChain-/tree/main/Assignment%202>